MATHEMATICAL ANALYSIS 2

TEST 3 - ANSWERS
Fall semester 2024/25

1. Consider the function g defined by

$$g(x,y) = y\sin(x) + x^2 + y^2$$

[1.5] (a) Prove that (0,0) is a critical point of g.

[2.5] (b) Classify the critical point (0,0) as a maximum, a minimum or a saddle point.

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(a) $\dfrac{\partial g}{\partial x} = y\cos(x) + 2x \quad$ and $\quad \dfrac{\partial g}{\partial y} = \sin(x) + 2y$

As $\dfrac{\partial g}{\partial x}(0,0) = 0$ and $\dfrac{\partial g}{\partial y}(0,0) = 0$, we conclude that (0,0) is a critical point of g.

(b) To classify the critical point (0,0), we need to study the sign of $d^2g(0,0)$.

$$H(x,y) = \begin{bmatrix} -y\sin(x) + 2 & \cos(x) \\ \cos(x) & 2 \end{bmatrix} \Rightarrow H(x,y) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Principal minors of $H(0,0)$:

$|H_1| = 2 \quad > 0$

$|H_2| = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3 \quad > 0$

As the principal minors of $H(0,0)$ are all positive, we conclude that $d^2g(0,0)$ is a positive definite quadratic form, that is $d^2g(0,0) > 0$. So, $g(0,0) = 0$ is a local minimum.

[3.0] 2. Consider the function f defined in $\mathbb{R}^2$ by:

$$f(x,y) = e^x\, y$$

Use Lagrange Multipliers Method to find the extreme values of f when subject to the constraint $x + y = 0$.

ANSWER

Consider the Lagrangean function:

$$L(\lambda, x, y) = e^x y + \lambda(0 - x - y)$$

First-order conditions: $\nabla L(\lambda, x, y) = (0,0,0) \Leftrightarrow$

$$\Leftrightarrow \begin{cases} \frac{\partial L}{\partial \lambda} = 0 \\ \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} -x - y = 0 \\ e^x y - \lambda = 0 \\ e^x - \lambda = 0 \end{cases} \Leftrightarrow \begin{cases} - \\ e^x y - e^x = 0 \\ \lambda = e^x \end{cases} \Leftrightarrow \begin{cases} - \\ e^x(y-1) = 0 \\ - \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} - \\ \underbrace{e^x = 0}_{False} \vee y = 1 \\ - \end{cases} \Leftrightarrow \begin{cases} x = -y \Leftrightarrow \boxed{x = -1} \\ \boxed{y = 1} \\ \boxed{\lambda = e^{-1}} \end{cases}$$

$(-1, 1)$ with $\lambda = e^{-1}$ is a candidate for problem solution.

Second-order conditions: Study the sign of the last (2-1=1) principal minor of the bordered Hessian matrix at $(e^{-1}, -1, 1)$.

$$\bar{H}(\lambda, x, y) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & e^x y & e^x \\ -1 & e^x & 0 \end{bmatrix} \Rightarrow \bar{H}(e^{-1}, -1, 1) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & e^{-1} & e^{-1} \\ -1 & e^{-1} & 0 \end{bmatrix}$$

$$\left|\bar{H}(e^{-1}, -1, 1)\right| = \begin{vmatrix} 0 & -1 & -1 \\ -1 & \frac{1}{e} & \frac{1}{e} \\ -1 & \frac{1}{e} & 0 \end{vmatrix} = (-1)(-1)^3 \begin{vmatrix} -1 & \frac{1}{e} \\ -1 & 0 \end{vmatrix} + (-1)(-1)^4 \begin{vmatrix} -1 & \frac{1}{e} \\ -1 & \frac{1}{e} \end{vmatrix} =$$

$$= 0 + \frac{1}{e} - \left(-\frac{1}{e} + \frac{1}{e}\right) = \frac{1}{e} \quad > 0$$

As the problem has 2 variables and 1 constraint, and the relevant principal minor is positive, we conclude that $f(-1, 1) = \dfrac{1}{e}$ is the maximum of f when subjected to the constraint $x+y = 0$.

3. Consider the following subset of $\mathbb{R}^2$:

$$A = \left\{(x,y) : y \leq \sqrt{x} \wedge y \geq x - 2 \wedge y \geq 0\right\}$$

[1.5] (a) Draw the region defined by the set A.

[2.0] (b) Using double integral(s) compute the area of A.

[3.0] (c) With a graphical approach and a detailed explanation, find the minimum and the maximum values of $h(x,y) = (x-4)^2 + (y-1)^2$ on the set A.

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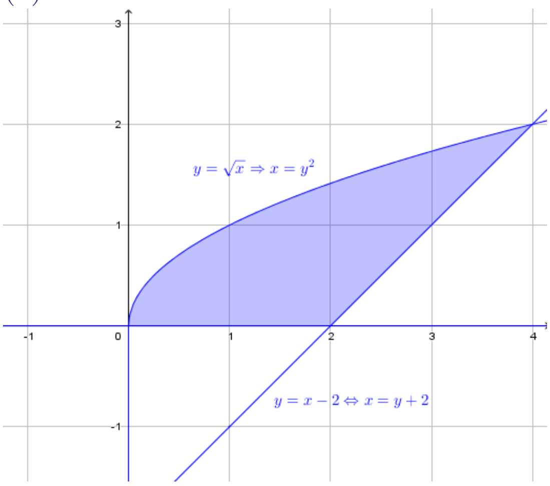
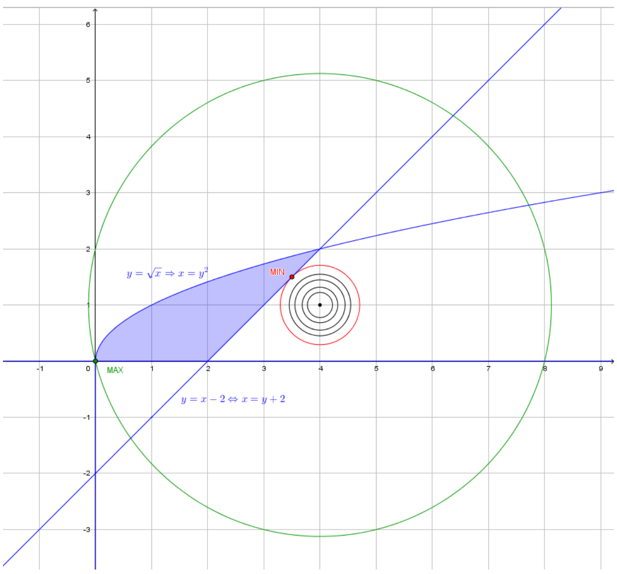
(a)



(b) $Area(A) = \iint_A 1 \; dxdy = \int_0^2 \int_{y^2}^{y+2} 1 \; dxdy = \int_0^2 [x]_{y^2}^{y+2} \; dy = \int_0^2 y + 2 - y^2 \; dy =$

$$= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_0^2 = 2 + 4 - \frac{8}{3} - 0 = \frac{18-8}{3} = \boxed{\frac{10}{3}}$$

(c) Consider the region drawn in (a) and draw level sets of h: $(x-4)^2 + (y-1)^2 = k$



To find the minimum we look for the "first" level set that intersects the region A. Analysing the graph, we can see that the minimum is at the tangency point between a level set and the line $\underbrace{y-x}_{g(x,y)}=-2$.

Calculus:

$$\begin{cases} \nabla h(x,y)=\lambda \nabla g(x,y) \\ g(x,y)=-2 \end{cases} \Leftrightarrow \begin{cases} (2(x-4),2(y-1))=\lambda(-1,1) \\ y-x=-2 \end{cases} \Leftrightarrow \begin{cases} 2x-8=-\lambda \\ 2y-2=\lambda \\ y-x=-2 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x=4-\dfrac{\lambda}{2} \\ y=1+\dfrac{\lambda}{2} \\ 1+\dfrac{\lambda}{2}-4+\dfrac{\lambda}{2}=-2 \end{cases} \Leftrightarrow \begin{cases} x=\dfrac{7}{2} \\ y=\dfrac{3}{2} \\ \lambda=1 \end{cases}$$

So, $h\left(\frac{7}{2},\frac{3}{2}\right)=\frac{1}{2}$ is the minimum of h on the set A.

To find the maximum we look for the "last" level set that intersects the region A. Analysing the graph we can see that the maximum is at the point (0,0). So, $h(0,0)=15$ is the maximum of h on the set A.

4. Consider the following double integral:

$$\int_0^1 \int_{-1}^{x} \frac{2xy}{(y^2+1)^2}\, dy\, dx$$

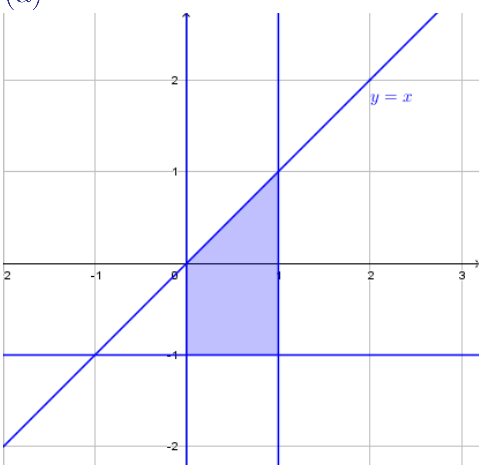
[1.5] (a) Draw the region of integration.

[2.0] (b) Interchange the order of integration.

[3.0] (c) Compute the given double integral.

ANSWER

(a)



(b) $\displaystyle\int_{-1}^{0}\int_{0}^{1}\frac{2xy}{(y^2+1)^2}\,dx\,dy+\int_{0}^{1}\int_{y}^{1}\frac{2xy}{(y^2+1)^2}\,dx\,dy$

(c) $\displaystyle\int_{0}^{1}\int_{-1}^{x}\frac{2xy}{(y^2+1)^2}\,dy\,dx=\int_{0}^{1}x\int_{-1}^{x}2y(y^2+1)^{-2}\,dy\,dx=\int_{0}^{1}x\left[\frac{(y^2+1)^{-1}}{-1}\right]_{-1}^{x}dx=$

$$=\int_{0}^{1}x\left[-\frac{1}{y^2+1}\right]_{-1}^{x}dx=\int_{0}^{1}x\left(-\frac{1}{x^2+1}+\frac{1}{2}\right)dx=-\frac{1}{2}\int_{0}^{1}\frac{2x}{x^2+1}\,dx+\int_{0}^{1}\frac{x}{2}\,dx=$$

$$=-\frac{1}{2}\left[\ln|x^2+1|\right]_0^1+\left[\frac{x^2}{4}\right]_0^1=-\frac{1}{2}(\ln 2-\ln 1)+\frac{1}{4}-0=\boxed{-\frac{1}{2}\ln 2+\frac{1}{4}}$$

End